Discrete Applied Mathematics 379 (2026) 605–612

Contents lists available at ScienceDirect

Discrete Applied Mathematics

journal homepage: www.elsevier.com/locate/dam

Maximum and minimum core payoffs for supermodular multisided assignment markets with local monotonicity☆

F. Javier Martínez-de-Albéniz [a,*], Carlos Rafels [a], Neus Ybern [b]

[a] Dept. de Matemàtica Econòmica, Financera i Actuarial and BEAT, Universitat de Barcelona, Av. Diagonal 690, 08034 Barcelona, Spain

[b] Dept. de Matemàtiques, Escola Politècnica Superior d'Enginyeria, Universitat Politècnica de Catalunya, Av. Víctor Balaguer 1, 08800 Vilanova i la Geltrú, Barcelona, Spain

ARTICLE INFO

Article history:
Received 17 January 2025
Received in revised form 9 September 2025
Accepted 1 October 2025

Keywords:
Supermodular market
Assignment game
Multisided assignment game
Core
Extreme core allocations

ABSTRACT

We give formulas to obtain the maximum and minimum core payoffs for supermodular multisided assignment markets satisfying a local monotonicity condition. These formulas are obtained directly from the multisided array of data and we provide a core allocation where these bounds are attained. Similar formulas are known from Eriksson et al. (2000) for the bilateral Becker's assortative assignment.

© 2025 Published by Elsevier B.V.

1. Introduction

Multisided assignment markets present an economic scenario where agents from at least two sides must be matched to generate profit, with one agent from each side participating. These sides represent various types of individuals or entities, such as buyers, sellers, lawyers, or skilled workers. The central challenge lies in distributing the profits resulting from the collective actions of these agents.

In the two-sided case, as demonstrated by Shapley and Shubik (1971) [9], the resulting assignment game is always balanced, ensuring that its core is non-empty. However, this significant finding does not hold true when dealing with three or more sides, as observed by Kaneko and Wooders (1982) [6] and Quint (1991) [8]. Consequently, different classes of balanced multisided games have been explored, including the works of Quint (1991) [8], Stuart (1997) [11], Sherstyuk (1999) [10], Tejada (2013) [12] or Atay and Núñez (2019) [2], all focusing on core-related questions.

Of relevance to our research is the work of Sherstyuk (1999) [10], which addresses the non-emptiness of the core in supermodular multisided assignment markets. Moreover, her paper focuses on assortative games that are both supermodular and monotonic across all array entries. In Eriksson et al. (2000) [5], formulas for obtaining the maximum and the minimum core payoffs are provided for the assortative case, but only in the two-sided scenario. The concept of supermodularity is central to our investigation and is justified and utilized in various economic applications. Notably, Topkis' (1998) [13] seminal book extensively covers supermodularity and related topics in both non-cooperative and cooperative games.

☆ Financial support from research grant PID2023-150472NB-I00 (Ministerio de Ciencia, Innovación y Universidades, AEI/FEDER, UE) is gratefully acknowledged, and from the Generalitat de Catalunya, through grant 2021 SGR 00306.

* Corresponding author.
E-mail address: javier.martinezdealbeniz@ub.edu (F.J. Martínez-de-Albéniz).

https://doi.org/10.1016/j.dam.2025.10.006
0166-218X/© 2025 Published by Elsevier B.V.

In this paper, we introduce a method for computing optimal core payoffs for all agents from each side directly from market data, eliminating the need for tedious calculations on the characteristic function of the multisided assignment game. While it is well-known that optimal payoffs for each side are achieved at the same core allocation in the bilateral case, in multisided assignment games, complexities arise since the efficiency of any core element is insufficient to determine the entire vector when only the payoffs of one side are known. Although Martínez-de-Albéniz et al. (2020) [1] presents a method to determine all the extreme core allocations for assortative multisided assignment games, a formula for obtaining optimal core payoffs is not provided.

Multisided market models offer insights into diverse economic situations involving various types of traders, consumers, and producers, and allow for comparisons between bilateral and multilateral cases. Here, we derive formulas applicable to both two-sided and multisided scenarios, imposing two conditions on the original array: supermodularity and monotonicity between certain entries, see condition (4). Supermodularity serves as a complementary condition on agents' abilities, ensuring, in the square case, that the main diagonal is always optimal and that the core is non-empty, facilitating the search for the optimum core payoffs.

Now it is time to highlight social and economic models that align with our underlying assumptions: supermodularity and monotonicity in multisided assignment markets. We focus on three models.

The first model that perfectly fits multisided assignment games with complementarities is Becker's model (Becker, 1973 [3]). Becker, a pioneering economist, extended economic analysis to various aspects of human behavior, including marriage. In his model, he examines how individuals choose partners based on certain traits or characteristics, leading to assortative mating. Assortative mating refers to the tendency of individuals to pair with others who have similar characteristics, such as education, income, or social status. Becker's model suggests that this behavior is rational, as individuals seek to maximize their utility through marriage. The basic hypothesis of this model departs from the supermodularity of the matrix and full monotonicity of its entries. Some recent works on assortative assignment markets include Eriksson et al. (2000) [5] and Martínez-de-Albéniz et al. (2020) [1].

Secondly, we have the famous Böhm-Bawerk markets, also known as horse markets. In the original horse market, various buyers and sellers come together, each with different valuations and expectations. Buyers have different maximum prices they are willing to pay, while sellers have different minimum prices they are willing to accept. These prices do not depend on who they are matched with. Through the process of negotiation and exchange, an equilibrium price emerges, reflecting the marginal utility of the horses to both buyers and sellers. As an assignment model, it was first studied by Shapley and Shubik (1971) [9] in the bilateral case, but in Tejada (2013) [12], the model is extended to multisided sectors. In this case each buyer needs an item from each different seller: one horse, one saddle, etc. This model is always supermodular and satisfies the monotonicity conditions.

Finally, we need to highlight the supermodular assignment markets generated by a production function. We evaluate teams formed by exactly one agent from each side of the market. Each agent has an initial endowment, and a general production function determines the worth of the teams in monetary terms. The goal is to match agents with different skills in a way that maximizes total production. More details on these multisided production models can be found in Eriksson et al. (2000) [5] (the bilateral case) and Burkard et al. (2009) [4] (the multilateral case). It is important to note that when the production function satisfies nonnegative second cross partial derivatives, a supermodular multisided assignment market is obtained. Introducing costs to the cooperation of the agents allows modeling the local conditions for these multisided assignment markets with costs.

The paper is structured as follows. Section 2 presents notations on multisided assignment games. Section 3 is dedicated to the main results and their proofs. Finally, Section 4 concludes and outlines future research directions.

2. Notations and terminology

We follow the notation used in Martínez-de-Albéniz et al. (2020) [1]. A *multisided assignment market* $(N^1, N^2, \dots, N^m; A)$ is formed by m non-empty pairwise disjoint finite sets of agents, $N^k = \left\{1^k, 2^k, \dots, n_k^k\right\}$ for $k \in M = \{1, \dots, m\}$ and a non-negative m-dimensional array $A = (a_E)_{E \in \Pi_{k=1}^m N^k}$. Any m-tuple of agents $E = (i_1, \dots, i_m) \in \Pi_{k=1}^m N^k$ is called an *essential coalition*.[1] Each set N^k is called a *side* and corresponds to one type of agents, such as different professional skills.

When the number of agents is the same in each side $|N^1| = |N^2| = \cdots = |N^m| = n$ the multisided assignment market is said to be n-square. Clearly, we can think of array A as a function f defined on the set $\Pi_{k=1}^m N^k$, that is, $a_E = f(E)$.

Shapley and Shubik (1971) [9], followed by Quint (1991) [8], associate any two-sided or multisided assignment market with a cooperative game, known as the *assignment game*, (N, w_A).[2] The set of players $N = \bigcup_{k=1}^m N^k$ consists of all agents involved and the characteristic function w_A is defined for each coalition based on the best outcome that its members can achieve. This involves partitioning the player set into essential coalitions and singletons to maximize their collective profit. The collection of essential coalitions that yields the optimal outcome for a given coalition is referred to as an

[1] To simplify notation, when no confusion arises, we will drop the superscript to describe the agents in N^k, i.e. $N^k = \{1, 2, \dots, n_k\}$. Its cardinality is $|N^k| = n_k$.

[2] In a cooperative game (N, v), the set of players is given by $N = \{1, \dots, n\}$ and v is a function that assigns a real number $v(S)$ for any coalition $S \subseteq N$ with $v(\emptyset) = 0$. Its core is defined as $C(v) := \{x \in \mathbb{R}^n \mid \sum_{i \in N} x_i = v(N)$ and for all $S \subseteq N$, $\sum_{i \in S} x_i \geq v(S)\}$. A game is named balanced if its core is non-empty.

optimal matching for that coalition. Formally, for any coalition $S \subseteq N$ a *matching* μ among $S \cap N^1, \ldots, S \cap N^m$ is a set of essential coalitions $\mu \subseteq \Pi_{k=1}^m (S \cap N^k)$ such that any agent of S belongs at most to a coalition in μ, and its cardinality is $|\mu| = \min_{k \in M} |S \cap N^k|$. The set of all matchings is denoted by $\mathcal{M}\left(S \cap N^1, \ldots, S \cap N^m\right)$. A matching μ is *optimal* if it maximizes $\sum_{E \in \mu} a_E$ over the set $\mathcal{M}\left(S \cap N^1, \ldots, S \cap N^m\right)$. The set of all optimal matchings for the grand coalition is denoted by $\mathcal{M}_A^*\left(N^1, \ldots, N^m\right)$. As a consequence, for any $\mu \in \mathcal{M}_A^*\left(N^1, \ldots, N^m\right)$ we have $w_A(N) = \sum_{E \in \mu} a_E$.

We focus on multisided assignment markets $(N^1, N^2, \ldots, N^m; A)$ which are *supermodular*,[3] that is

$$a_E + a_{E'} \leq a_{E \vee E'} + a_{E \wedge E'} \quad \text{for all } E, E' \in \Pi_{k=1}^m N^k. \tag{1}$$

The agents of a multisided assignment game may divide among themselves their worth, $w_A(N)$, in any way they like. Thus, an *allocation* is a non-negative vector $x = (x^1, x^2, \ldots, x^m) \in \Pi_{k=1}^m \mathbb{R}_+^{n_k}$ such that $\sum_{k=1}^m \sum_{i_k=1}^{n_k} x_{i_k}^k = w_A(N)$. Vector $x^k \in \mathbb{R}_+^{n_k}$ is interpreted as the payoffs to agents in N^k, i.e. x_i^k is the payoff associated to player i^k, that is player i of side k. For any essential coalition $E = (i_1, \ldots, i_m) \in \Pi_{k=1}^m N^k$ we write $x(E) = \sum_{k=1}^m x_{i_k}^k$.

The *core of the multisided assignment game* $C(w_A)$ is described for any fixed optimal matching $\mu \in \mathcal{M}_A^*\left(N^1, \ldots, N^m\right)$ as those allocations $x \in \Pi_{k=1}^m \mathbb{R}_+^{n_k}$ satisfying

$$x(E) = a_E \quad \text{for all } E \in \mu,$$
$$x(E) \geq a_E \quad \text{for all } E \notin \mu,$$

and unassigned agents by μ receive a zero payoff in any core allocation.

From now on, we concentrate in the n-square case. We denote by

$$E_{\boldsymbol{i}} = (i, \ldots, i), \text{ for } i = 1, 2, \ldots, n,$$

the essential coalitions of the main diagonal. Notice that $\mu = \{E_{\boldsymbol{1}}, E_{\boldsymbol{2}}, \ldots, E_{\boldsymbol{n}}\}$ is called the main diagonal matching.

The main diagonal matching is optimal in the n-square supermodular case, maybe not the only optimal matching.[4] Moreover for supermodular multisided markets (see Sherstyuk, 1999, [10]) it is known that the core is always non-empty, $C(w_A) \neq \varnothing$.

In Martínez-de-Albéniz et al. (2020) [1] it is proved (Theorem 2.1 and Remark 2.1) that for n-square supermodular multisided markets only the central strip of array A is necessary to determine the core conditions. This central strip is formed by all $E = (i_1, i_2, \ldots, i_m) \in \Pi_{k=1}^m N^k$ such that $E_{\boldsymbol{i-1}} \leq E \leq E_{\boldsymbol{i}}$ for $i = 2, \ldots, n$. It is equivalent to $\max_{k \in M} i_k - \min_{k \in M} i_k \leq 1$. Its core $C(w_A)$ can be described as those vectors $x \in \Pi_{k=1}^m \mathbb{R}_+^{n_k}$ satisfying

$$x(E_{\boldsymbol{i}}) = a_{i \ldots i} \quad \text{for all } i = 1, \ldots, n, \tag{2}$$
$$x(E) \geq a_E \quad \text{for all } E \text{ in the central strip.} \tag{3}$$

We ask that our square multisided assignment markets $(N^1, N^2, \ldots, N^m; A)$ satisfy an additional *local monotonicity condition*:

$$a_{ii \cdots i} \leq a_{i \cdots i \underbrace{i+1}_{k)} i \cdots i}, \quad \text{for all } i = 1, \ldots, n-1 \text{ and all } k = 1, \ldots, m. \tag{4}$$

This local monotonicity condition here is different from the assortative condition used in Martínez-de-Albéniz et al. (2020) [1], Eriksson et al. (2000) [5] or Sherstyuk (1999) [10] which asks for a monotonicity for all entries, that is

$$a_E \leq a_{E'} \quad \text{for all } E \leq E', \quad E, E' \in \Pi_{k=1}^m N^k. \tag{5}$$

The assortative condition (5) implies the local monotonicity (4), but the reverse is not true, as the following example shows.

Example 2.1. Consider the following numerical $3 \times 3 \times 3$ multisided assignment game:

$$A = \begin{pmatrix} a_{111} & a_{121} & a_{131} \\ a_{211} & a_{221} & a_{231} \\ a_{311} & a_{321} & a_{331} \end{pmatrix} \begin{pmatrix} a_{112} & a_{122} & a_{132} \\ a_{212} & a_{222} & a_{232} \\ a_{312} & a_{322} & a_{332} \end{pmatrix} \begin{pmatrix} a_{113} & a_{123} & a_{133} \\ a_{213} & a_{223} & a_{233} \\ a_{313} & a_{323} & a_{333} \end{pmatrix} =$$
$$\begin{pmatrix} \mathbf{2} & 6 & 2 \\ 4 & 11 & 8 \\ 0 & 8 & 6 \end{pmatrix} \begin{pmatrix} 6 & 10 & 9 \\ 8 & \mathbf{15} & 17 \\ 8 & 16 & 20 \end{pmatrix} \begin{pmatrix} 3 & 7 & 10 \\ 7 & 15 & 19 \\ 7 & 19 & \mathbf{100} \end{pmatrix}.$$

This is a supermodular array and then its main diagonal is optimal (in boldface). Moreover, it satisfies the local monotonicity condition (4) but not the assortative condition (5) since, for example, $a_{121} = 6 \not\leq a_{131} = 2$.

[3] For any pair of essential coalitions $E, E' \in \Pi_{k=1}^m N^k$ we define $E \vee E'$ as another essential coalition which elements are the maximum component-wise and $E \wedge E'$ as the corresponding minimum component-wise.

[4] Notice that if there are two essential coalitions E, E' of μ that are not comparable, we can use supermodularity to obtain a new optimal matching with $E \vee E'$ and $E \wedge E'$, two essential coalitions now comparable.

Now we show an example of a multisided market generated by a production function with costs, following the model outlined in the introduction.

Example 2.2 (*A Multisided Production Market with Costs*)**.** Consider a production function with three types of agents and three individuals on each side. The production function is given by a Cobb–Douglas function: $f(x, y, z) = (xyz)^{1/3}$, where x, y and z represent the inputs of the agents of the first, second, and third types, respectively. The endowments of the agents are as follows: for the first type $a = (8, 216, 1000)$; for the second type, $b = (27, 125, 512)$; and for the third type $c = (64, 343, 729)$. We can obtain the production simply by combining the initial endowments.

The tax of c units for those essential coalitions when the distance between the members of an essential coalition is at least 2. Being neighbors is better for business. Define for an essential coalition $E = (i_1, i_2, i_3)$ the rank of the coalition as

$$r(E) = \max_{k\in\{1,2,3\}} i_k - \min_{k\in\{1,2,3\}} i_k,$$

and the cost is applied when $r(E) \geq 2$. Now we define the entries of this production multisided assignment array with costs as:

$$a_{ijk} = \begin{cases} f(a_i, b_j, c_k) & \text{if } r(i, j, k) \leq 1, \\ f(a_i, b_j, c_k) - c \cdot r(i, j, k) & \text{if } r(i, j, k) \geq 2. \end{cases}$$

The following array is deduced from the above data, with an unitary cost of $c = 27$.

$$A = \begin{pmatrix} \mathbf{24} & 40 & 10 \\ 72 & 120 & 138 \\ 66 & 146 & 266 \end{pmatrix} \quad \begin{pmatrix} 42 & 70 & 58 \\ 126 & \mathbf{210} & 336 \\ 156 & 350 & 560 \end{pmatrix} \quad \begin{pmatrix} 0 & 36 & 90 \\ 108 & 270 & 432 \\ 216 & 450 & \mathbf{720} \end{pmatrix}.$$

This is a supermodular array satisfying local monotonicity, but not full monotonicity.

3. Main results

In this section we provide explicit formulas for both the maximum and minimum core payoffs, achievable for any agent in a multisided supermodular n-square assignment market satisfying the local monotonicity condition (4). Additionally we also show that all agents belonging to the same side attain their maximum at the same core allocation, and similarly for the minimum.

Let $(N^1, N^2, \ldots, N^m; A)$ be a n-square multisided supermodular assignment market, and since its core is a non-empty and compact set, for each agent there is a maximum and minimum core payoff. The maximum/minimum core payoff for agent i^k, are denoted by M_i^k, and m_i^k, respectively:

$$M_i^k = \max_{x\in C(w_A)} x_i^k, \quad \text{and} \quad m_i^k = \min_{x\in C(w_A)} x_i^k.$$

For notational convenience to allow for compact formulas, whenever in a formula index 0 appears, we assume that the corresponding array element is zero.

Our first theorem establishes that a particular vector belongs to the core. We will prove later that in this vector we attain the maximum core payoff for all agents of the first side, and the minimum core payoff for all agents of the last side. We focus primarily on the first side, and for any other side we simply reorder the sides accordingly.[5]

Theorem 3.1. *Let $(N^1, N^2, \ldots, N^m; A)$ be a n-square multisided supermodular assignment market satisfying the local monotonicity condition (4). Then, the vector $x = \left(x_i^k\right)_{i=1,\ldots,n}^{k=1,\ldots,m}$ defined by:*

$$x_i^k = \sum_{r=1}^{i} \left(a_{\underbrace{r-1\cdots r-1}_{k-1} r\cdots r} - a_{\underbrace{r-1\cdots r-1}_{k} r\cdots r} \right), \text{ for } i = 1, \ldots, n \text{ and } k = 1, \ldots, m, \tag{6}$$

belongs to the core $C(w_A)$.

Proof. Its proof can be found in Appendix. □

The following example shows how to find this vector x.

Example 3.1 (*Example 2.1 Revisited*)**.** We revise Example 2.1 and we apply Theorem 3.1 to find the following vector:

$$x = \left(x_1^1, x_2^1, x_3^1; x_1^2, x_2^2, x_3^2; x_1^3, x_2^3, x_3^3\right),$$

where the entries can be found as follows:

$$x_1^1 = a_{111} = 2, \qquad x_2^1 = a_{111} + (a_{222} - a_{122}) = 7,$$

[5] Notice that permuting two sides preserves the supermodularity of the corresponding assignment market array and also satisfies condition (4). For example, by exchange of side 1 with side k we obtain a new array $\widehat{A}$ where $\widehat{a}_{i_1\ldots i_k\ldots i_m} = a_{i_k i_2\ldots i_{k-1} i_1 i_{k+1}\ldots i_m}$.

$$x_3^1 = a_{111} + (a_{222} - a_{122}) + (a_{333} - a_{233}) = 88;$$
$$x_1^2 = 0, \qquad x_2^2 = 0 + (a_{122} - a_{112}) = 4,$$
$$x_3^2 = 0 + (a_{122} - a_{112}) + (a_{233} - a_{223}) = 8;$$
$$x_1^3 = 0, \qquad x_2^3 = 0 + (a_{112} - a_{111}) = 4,$$
$$x_3^3 = 0 + (a_{112} - a_{111}) + (a_{223} - a_{222}) = 4.$$

Then we have $x = (2, 7, 88; 0, 4, 8; 0, 4, 4)$, which is a core allocation of (N, w_A).

Our next theorem gives the expressions for the maximum and the minimum core payoffs in terms of the array's entries.

Theorem 3.2. *Let $(N^1, N^2, \ldots, N^m; A)$ be a n-square multisided supermodular assignment market satisfying the local monotonicity condition (4). Then, the maximum core payoff for agent i^k is given by:*

$$M_i^k = \sum_{r=1}^{i} \left(a_{rr\cdots r} - a_{rr\cdots r\underbrace{r-1\,r\cdots r}_{k)}} \right), \tag{7}$$

for $i = 1, \ldots, n$ and $k = 1, \ldots, m$.

The minimum core payoff for agent i^k is given by:

$$m_i^k = \sum_{r=1}^{i} \left(a_{r-1\cdots r-1\underbrace{r}_{k)}r-1\cdots r-1} - a_{r-1\cdots r-1} \right), \tag{8}$$

for $i = 1, \ldots, n$ and $k = 1, \ldots, m$.

Moreover, all these maxima/minima core payoffs for all agents on the same side are attained at the same core allocation.

Proof. First, we show that for any core element $x \in C(w_A)$ we have $x_i^k \le M_i^k$, for all $i = 1, \ldots, n$, and $k = 1, \ldots, m$. Since $\sum_{k=1}^m x_1^k = a_{11\cdots1}$, and the vector is positive, we have $x_1^k \le M_1^k = a_{11\cdots1}$, for $k = 1, \ldots, m$. By (2), $\sum_{l=1}^m x_r^l = a_{rr\cdots r}$ and by (3), $\sum_{l=1}^{k-1} x_r^l + x_{r-1}^k + \sum_{l=k+1}^m x_r^l \ge a_{rr\cdots r\underbrace{r-1\,r\cdots r}_{k)}}$. Then for $r = 2, \ldots, n$, we have $x_r^k - x_{r-1}^k \le a_{rr\cdots r} - a_{rr\cdots r\underbrace{r-1\,r\cdots r}_{k)}}$ for $k = 1, \ldots, m$.

By summing up all these inequalities we obtain

$$x_i^k = \sum_{r=1}^{i} \left(x_r^k - x_{r-1}^k\right) \le \sum_{r=1}^{i} \left(a_{rr\cdots r} - a_{rr\cdots r\underbrace{r-1\,r\cdots r}_{k)}} \right) = M_i^k,$$

for all $i = 1, \ldots, n$ and $k = 1, \ldots, m$.

Now we show that for any $x \in C(w_A)$ we have $x_i^k \ge m_i^k$, for all $i = 1, \ldots, n$, and $k = 1, \ldots, m$.

Let $x \in C(w_A)$. Since the vector is positive and $m_1^k = 0$ for $k = 1, \ldots, m$, we have $x_1^k \ge 0 = m_1^k$, for $k = 1, \ldots, m$. By (2), $\sum_{l=1}^m x_{r-1}^l = a_{r-1\cdots r-1}$ and by (3), $\sum_{l=1}^{k-1} x_{r-1}^l + x_r^k + \sum_{l=k+1}^m x_{r-1}^l \ge a_{r-1\cdots r-1\underbrace{r}_{k)}r-1\cdots r-1}$. Then for $r = 2, \ldots, n$, we have $x_r^k - x_{r-1}^k \ge a_{r-1\cdots r-1\underbrace{r}_{k)}r-1\cdots r-1} - a_{r-1\cdots r-1}$ for $k = 1, \ldots, m$.

By summing up all these inequalities we achieve

$$x_i^k = \sum_{r=1}^{i} \left(x_r^k - x_{r-1}^k\right) \ge \sum_{r=1}^{i} \left(a_{r-1\cdots r-1\underbrace{r}_{k)}r-1\cdots r-1} - a_{r-1\cdots r-1} \right) = m_i^k,$$

for all $i = 1, \ldots, n$ and $k = 1, \ldots, m$.

Now we show that the previously defined upper and lower bounds, M_i^k and m_i^k, for all $i = 1, \ldots, n$ and $k = 1, \ldots, m$, can indeed be attained within the core of the multisided assignment game. To reach this result, we focus on the first side and consider the vector x introduced in Theorem 3.1. In this vector the maximum payoff for all agents of the first side is achieved. Indeed, notice that for the first side we have: $x_i^1 = \sum_{r=1}^i a_{r\cdots r} - \sum_{r=1}^i a_{r-1\,r\cdots r} = M_i^1$, for $i = 1, \ldots, n$. That is, in this vector x, we attain the upper bound for all agents of the first side. Also we obtain the minimum core payoff for all agents of side m is achieved. Indeed, notice that for the last side we have:

$$x_i^m = \sum_{r=1}^{i} \left(a_{\underbrace{r-1\cdots r-1\,r}_{m-1}} - a_{r-1\cdots r-1} \right) = m_i^m, \text{ for } i = 1, \ldots, n.$$

That is, in this vector x, we attain the lower bound for all agents on the last side. □

Remark 3.1. Notice that the maximum and minimum core payoff can be described in a recursive way:

$$M_i^k = M_{i-1}^k + \left(a_{ii\cdots i} - a_{ii\cdots i\underbrace{i-1\ i\cdots i}_{k)}} \right),$$

with $M_0^k = 0$, for $i = 1, \ldots, n$ and $k = 1, \ldots, m$, and

$$m_i^k = m_{i-1}^k + \left(a_{i-1\cdots i-1\underbrace{i\ i-1\cdots i-1}_{k)}} - a_{i-1\cdots i-1} \right),$$

with $m_0^k = 0$, for $i = 1, \ldots, n$ and $k = 1, \ldots, m$.

These formulas show how we move from the maximum or minimum payoff of one agent to the next. Differences in parentheses represent the increase in payoffs between two consecutive agents.

Remark 3.2. In Theorem 3.2 we have proved that the maximum core payoffs for all agents from the same side are attained at the same core allocation. The vector defined in (6) gives the expression for agents on the first side. However, this core allocation may not be unique, and this is a strong difference with two-sided assignment markets, where this core allocation is always unique. We present a numerical example, Example 3.2, to illustrate this point.

Example 3.2. Consider the following $2 \times 2 \times 2$ multisided assignment market:

$$A = \begin{pmatrix} a_{111} & a_{121} \\ a_{211} & a_{221} \end{pmatrix} \quad \begin{pmatrix} a_{112} & a_{122} \\ a_{212} & a_{222} \end{pmatrix} = \begin{pmatrix} \mathbf{2} & 3 \\ 4 & 6 \end{pmatrix} \quad \begin{pmatrix} 5 & 7 \\ 8 & \mathbf{10} \end{pmatrix}.$$

The main diagonal is optimal and its elements are in boldface. The array is supermodular and satisfies the local monotonicity condition (4). We present the vector x defined in (6):

$$x = (x_1^1, x_2^1; x_1^2, x_2^2; x_1^3, x_2^3) = (2, 5; 0, 2; 0, 3).$$

By Theorem 3.1 the vector x is a core allocation. Furthermore, the first side receives its maximum core payoffs: $x_1^1 = M_1^1 = a_{111}$ and $x_2^1 = M_2^1 = a_{111} + (a_{222} - a_{122})$.

However, the vector $y = (2, 5; 0, 1; 0, 4)$ is also a core allocation, and agents of the first side get the same payoffs as in the vector x. Therefore, the vector at which the maximum is attained may not be unique. This argument also applies to the minimum.

Our next example demonstrates the necessity of the local monotonicity assumption, as the maximum is not provided by formula (7) if it fails.

Example 3.3. Consider the following numerical $2 \times 2 \times 2$ multisided assignment game:

$$A = \begin{pmatrix} \mathbf{4} & 3 \\ 6 & 5 \end{pmatrix} \quad \begin{pmatrix} 5 & 4 \\ 7 & \mathbf{6} \end{pmatrix}.$$

The array is supermodular and does not satisfy the local monotonicity condition (4), since $a_{111} = 4 > a_{121} = 3$. The maximum core payoffs given in Theorem 3.2 for the first side are $M_1^1 = a_{111} = 4$ and $M_2^1 = a_{111} + a_{222} - a_{122} = 6$, but there is no core element with these payoffs for the first side.

4. Concluding remarks

It should be noted that the formulas obtained in Theorem 3.2 can be applied to bilateral assignment games. As a result, our contributions extend the work of Eriksson et al. (2000) [5], which focuses solely on the bilateral case. Furthermore, we have relaxed the monotonicity condition by shifting from the stringent requirement of total monotonicity to the more flexible local monotonicity condition (4). The proofs for our multisided case are more complex than those of Eriksson et al. (2000) [5]. In the bilateral case, fixing the payoffs for one side completely determines the payoffs for the other side. In contrast, our approach requires discovering a new core allocation that serves all sides. Looking ahead, a key objective for future research is to propose algorithms that identify all extreme core allocations in supermodular multisided assignment markets satisfying local monotonicity. Developing such algorithms could open the way for more detailed analyses of specific solution concepts, such as the Shapley value, the tau-value, the nucleolus, etc., and their interrelationships (see Nuñez and Rafels, 2015 [7]). Moreover, having a simple and computable formula for the minimum and maximum payoff vectors for each side would be highly beneficial. This could aid in deriving comparative statics results or designing allocation mechanisms that are partially strategy-proof for multisided assignment markets.

Appendix. Proof of Theorem 3.1

Proof. Recall the definition of vector $x = \left(x_i^k\right)_{i=1,\dots,n}^{k=1,\dots,m}$ as follows:

$$x_i^k = \sum_{r=1}^{i} \left(a_{\underbrace{r-1\cdots r-1}_{k-1} r\cdots r} - a_{\underbrace{r-1\cdots r-1}_{k} r\cdots r} \right), \text{ for } i = 1,\dots,n \text{ and } k = 1,\dots,m.$$

First, we prove that $x \geq 0$ and it satisfies condition (2).

To prove the non-negativity of vector x, we show that $a_{\underbrace{r-1\cdots r-1}_{k-1} r\cdots r} - a_{\underbrace{r-1\cdots r-1}_{k} r\cdots r} \geq 0$ for all $r = 1,\dots,n$, and $k = 1,\dots,m$.

To achieve this, for any essential coalition formed only by agents $i-1$ and i with $i = 2,\dots,n$ replacing an agent $i-1$ by an agent i, results in an increase in the corresponding array entry.

We denote by $E_{\boldsymbol{i}}^S = (i-1)\mathbf{1}_{M\setminus S} + i\mathbf{1}_S$ where $S \subseteq M$, and agent $i = 2,\dots,n$.[6] We prove that for all $S \subseteq M$,

$$a_{E_i^{S\cup\{k\}}} - a_{E_i^S} \geq 0, \tag{A.1}$$

for all $i = 2,\dots,n$, and all $k \in M \setminus S$.

Our argument to prove (A.1) proceeds by induction on the cardinality s of coalition S. For $s = 0$, that is $S = \varnothing$, these inequalities (A.1) reduce to local monotonicity. Assume the inequalities in (A.1) have been proved for coalitions of size $s-1$. We show they also hold for coalitions of size s.

Without loss of generality we take coalition $S \subseteq M$ to be the last s sides of the market, $s \geq 1$. By supermodularity,

$$a_{i-1\cdots i-1\underbrace{i\cdots i}_{s-1}} + a_{i-1\cdots i-1\underbrace{i\cdots i}_{s+1}} \geq a_{i-1\cdots i-1\underbrace{i\cdots i}_{s}} + a_{\underbrace{i-1\cdots i-1}_{m-s-1} i\, i-1 \underbrace{i\cdots i}_{s-1}},$$

or equivalently:

$$a_{i-1\cdots i-1\underbrace{i\cdots i}_{s+1}} - a_{i-1\cdots i-1\underbrace{i\cdots i}_{s}} \geq a_{\underbrace{i-1\cdots i-1}_{m-s-1} i\, i-1 \underbrace{i\cdots i}_{s-1}} - a_{i-1\cdots i-1\underbrace{i\cdots i}_{s-1}} \geq 0,$$

where the last inequality holds by induction hypothesis.

To check (2), notice for $i = 1,\dots,n$:

$$\begin{aligned}
\sum_{k=1}^{m} x_i^k &= \sum_{k=1}^{m} \left(\sum_{r=1}^{i} \left(a_{\underbrace{r-1\cdots r-1}_{k-1} r\cdots r} - a_{\underbrace{r-1\cdots r-1}_{k} r\cdots r} \right) \right) \\
&= \sum_{r=1}^{i} \left(\sum_{k=1}^{m} \left(a_{\underbrace{r-1\cdots r-1}_{k-1} r\cdots r} - a_{\underbrace{r-1\cdots r-1}_{k} r\cdots r} \right) \right) \\
&= \sum_{r=1}^{i} \left(a_{rr\cdots r} - a_{r-1\,\dots,r-1} \right) \quad = a_{ii\cdots i}.
\end{aligned}$$

Now we prove that vector x satisfies the core inequalities given in (3). We initially claim that vector x satisfies some of the entries of the array with equality. We denote by $E_{\boldsymbol{i}}^s$ for $i = 2,\dots,n$, and $1 \leq s \leq m$, the essential coalitions

$$E_{\boldsymbol{i}}^s = (\underbrace{i-1, i-1, \dots, i-1}_{s}, i, \dots, i). \tag{A.2}$$

Now we justify that vector x satisfies the core constraints related to the above essential coalitions in (A.2) with equality. Therefore we must prove that

$$x(E_{\boldsymbol{i}}^s) = a_{E_{\boldsymbol{i}}^s} \text{ for } i = 2,\dots,n, \quad \text{and } 1 \leq s \leq m. \tag{A.3}$$

It is easy to prove

$$\sum_{k=1}^{s} x_{i-1}^k = \sum_{r=1}^{i-1} a_{rr\cdots r} - \sum_{r=1}^{i-1} a_{\underbrace{r-1\cdots r-1}_{s} r\cdots r},$$

and

$$\sum_{k=s+1}^{m} x_i^k = \sum_{r=1}^{i} a_{\underbrace{r-1\cdots r-1}_{s} r\cdots r} - \sum_{r=1}^{i} a_{r-1\cdots r-1}.$$

[6] The indicator vector $\mathbf{1}_A \in \mathbb{R}^M$ is defined for any subset $A \subseteq M$, by $\mathbf{1}_A(x) = 1$ if $x \in A$ and $\mathbf{1}_A(x) = 0$ if $x \notin A$.

Therefore, we obtain

$$x(E_{\boldsymbol{i}}^s) = \sum_{k=1}^{s} x_{i-1}^k + \sum_{k=s+1}^{m} x_i^k = a_{\underbrace{i-1\cdots i-1}_{s} i\cdots i} = a_{E_{\boldsymbol{i}}^s}.$$

Finally we prove that $x(E) \geq a_E$ for all essential coalitions $E = (i_1, i_2, \ldots, i_m)$ with $\max_{k\in M} i_k - \min_{k\in M} i_k = 1$. We will prove it by induction over

$$t(E) = |\{k \in M \mid i_k = i\}|,$$

which represents the number of agents i in the essential coalition E. Clearly $1 \leq t(E) \leq m-1$, for any essential coalition satisfying $\max_{k\in M} i_k - \min_{k\in M} i_k = 1$.

If $t(E) = 1$ there is only one agent i in E with all other agents being $i-1$. If this agent i is placed in the last position of E we know that $x(E) = a_E$, see (A.3). If this agent i is placed in position l where $1 \leq l < m$, we have $E = E_i^{\{l\}} = (i-1, \ldots, i-1, \underbrace{i}_{l}, i-1, \ldots, i-1)$. Now take the essential coalition $E_{\boldsymbol{i}}^l = (\underbrace{i-1, i-1, \ldots, i-1}_{l}, i, \ldots, i)$. Directly, from its definition we have

$$E \wedge E_{\boldsymbol{i}}^l = (i-1, i-1, \ldots, i-1) = E_{\boldsymbol{i-1}}, \quad \text{and}$$
$$E \vee E_{\boldsymbol{i}}^l = (\underbrace{i-1, \ldots, i-1}_{l-1}, i, \ldots, i) = E_{\boldsymbol{i}}^{l-1}.$$

Now, using the supermodularity condition, we obtain

$$a_E + a_{E_{\boldsymbol{i}}^l} \leq a_{E_{\boldsymbol{i-1}}} + a_{E_{\boldsymbol{i}}^{l-1}},$$

and then

$$a_E \leq x(E_{\boldsymbol{i-1}}) + x(E_{\boldsymbol{i}}^{l-1}) - x(E_{\boldsymbol{i}}^l) = x(E).$$

This establishes our assertion for cases where the essential coalition in the central strip contains only one agent i, i.e., $t(E) = 1$.

Assume now that $x(E) \geq a_E$ for all essential coalitions in the central strip, i.e., with $\max_{k\in M} i_k - \min_{k\in M} i_k = 1$, such that the number of agents i in E is $t(E) \leq t-1$. Let E be an essential coalition satisfying $E_{\boldsymbol{i-1}} < E < E_{\boldsymbol{i}}$, with $t(E) = t$, i.e. the essential coalition E has t agents i.

Let l, $1 \leq l \leq m$ be the first side in E with an agent i. Take the essential coalition $E_{\boldsymbol{i}}^l = (\underbrace{i-1, \ldots, i-1}_{l}, i, \ldots, i)$. Directly from its definition the essential coalition $E \wedge E_{\boldsymbol{i}}^l$ satisfies the condition of being in the central strip such that $t(E \wedge E_{\boldsymbol{i}}^l) = t-1 < t$, and

$$E \vee E_{\boldsymbol{i}}^l = (\underbrace{i-1, \ldots, i-1}_{l-1}, i, i, \ldots, i) = E_{\boldsymbol{i}}^{l-1}.$$

By the supermodularity condition, we can conclude that $a_E + a_{E_{\boldsymbol{i}}^l} \leq a_{E\wedge E_{\boldsymbol{i}}^l} + a_{E_{\boldsymbol{i}}^{l-1}}$, and by induction on t, we obtain the desired inequality, $x(E) \geq a_E$ finishing the proof of our assertion and also the proof of the theorem. □

Data availability

No data was used for the research described in the article.

References

[1] F.J. Martínez-de Albéniz, C. Rafels, N. Ybern, Assortative multisided assignment games: The extreme core points, Games Econom. Behav. 120 (2020) 144–153.
[2] A. Atay, M. Núñez, Multi-sided assignment games on m-partite graphs, Ann. Oper. Res. 279 (2019) 271–290.
[3] G.S. Becker, A theory of marriage: Part i, J. Political Econ. 81 (4) (1973) 813–846.
[4] R.E. Burkard, M. Dell'Amico, S. Martello, Assignment Problems, Philadelphia, Society for Industrial and Applied Mathematics, 2009.
[5] K. Eriksson, J. Karlander, L.-E. Öller, Becker's assortative assignments: stability and fairness, Math. Social Sci. 39 (2000) 109–118.
[6] M. Kaneko, M.H. Wooders, Cores of partitioning games, Math. Social Sci. 3 (4) (1982) 313–327.
[7] M. Núñez, C. Rafels, A survey on assignment markets, J. Dyn. Games 3&4 (2015) 227–256.
[8] T. Quint, The core of an m-sided assignment game, Games Econom. Behav. 3 (1991) 487–503.
[9] L.S. Shapley, M. Shubik, The assignment game I: the core, Int. J. Game Theory 1 (1971) 111–130.
[10] K. Sherstyuk, Multisided matching games with complementarities, Int. J. Game Theory 28 (1999) 489–509.
[11] H.W. Stuart, The supplier–firm–buyer game and its m-sided generalization, Math. Social Sci. 34 (1) (1997) 21–27.
[12] O. Tejada, Analysis of the core of multisided Böhm-Bawerk assignment markets, TOP 21 (1) (2013) 189–205.
[13] D.M. Topkis, Supermodularity and Complementarity, Princeton University Press, 1998.